ST. MARY'S DSG KLOOF

PHYSICAL SCIENCES PAPER 1 (Physics)

GRADE 10 **TIME: 2 HOURS + 10 min reading**
November 2024 **MARKS: 135**

Examiner: K. Bolton **Moderators: G. Main and T. Kirsten**
NAME: _____ **MEMO** _____

Physical Sciences teacher: _____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of **20 pages**. Please check that your question paper is complete and write your name on it. A **Data sheet** is also provided.

2. Read the questions carefully.

3. It is in your own interest to write legibly and to present your work neatly.

4. Answer ALL questions on the paper.

5. **All working must be shown** and units may be used in substitutions, but **MUST** be given with answers. Work to 2 decimal places or significant figures where appropriate. Make sure your calculator is in the correct mode.

6. Some extra space on page 20 is included at the end of the question paper. If you run out of space for a question, use this extra space. Clearly indicate the question number of your answer, should you use this extra space.

Question	1	2	3	4	5	6	7	8	Total
Your mark									
Mark	12	15	9	8	15	38	15	23	135

QUESTION 1 – MULTIPLE CHOICE

1.1	A	B	C	D ✗
1.2	A	B	C ✗	D
1.3	A	B ✗	C	D
1.4	A	B	C ✗	D
1.5	A ✗	B	C	D
1.6	A	B	C	D ✗

[6x2=12]

QUESTION 2

2.1 ONE of A/H/C/E/G ✓ (1)

2.2 B AND F/C AND G/A AND E ✓✓ (2)

2.3 B/F✓ (1)

2.4.1 8 X 0,4 ✓= 3,2 s✓ (2)

2.4.2 25/2✓ = 1,25 cm✓ (2)

2.4.3 T = 0,4 s✓ (1)

2.4.4 0,68/2 ✓= 0,34 m✓ (2)

2.4.5 $f = \frac{1}{T}$

$= \frac{1}{4}$✓

= 2,5 Hz✓ (2)

2.4.6 v = fλ

= (2,5)(0,34) coe 2.4.4. and 2.4.5✓

= 0,85 $m.s.^{-1}$✓ (2)

[15]

QUESTION 3

3.1 ANY TWO ✓✓
All travel at speed of light.
All are transverse waves
All can travel in a vacuum
All can be diffracted,refracted,interference and polarised (2)

3.2 A = infrared ✓
B= X-rays ✓ (2)

3.3 UV ✓ (1)

3.4 Microwaves ✓ (1)

3.5 $c = f\lambda$
$3x\ 10^8 = f\ (0,2)$

$f= 1,5 \times 10^9$ Hz ✓

$E = hf$

$= (6,6 \times {}^{10}\text{-}34)$ ✓ $(1,5 \times 10^9)$

$= 9,9 \times 10^{-25}$ J✓ (3)

[9]

QUESTION 4

4.1 A region in space ✓where a magnetic object experiences a force✓ (1)

4.2 Soft/temporary magnet ✓ (1)

4.3 Like poles✓ (1)

4.4 D to C. ✓D = North pole as repels N of compass needle✓ (2)

4.5 Give a reason why the Earth's magnetic field is important to our existence. (2)

Protects ✓against solar flares/storms✓

[8]

QUESTION 5

5.1.1 Gained electrons ✓ (1)

5.1.2 Electrons in bead repelled from the surface✓
Leaving it positive✓
Opposites attract✓ (3)

5.1.3 Bead moves away from ruler✓ as now it is negative ✓ (i.e. repelled) (2)

5.2

5.2.1 Charge cannot be created or destroyed. ✓ Number of electrons lost = number of electrons gained✓ (2)

5.2.2 C to B✓ (1)

5.2.3 Qnew = $\frac{4\ x\ 10^{-6}+(-6\ x\ 10^{-6})}{2}$✓
= - 1 x 10^{-6} C✓ (2)

5.2.4 Number of electrons = $\frac{-1\ x\ 10^{-6}-\ 4\ x10^{-6}}{-1{,}6\ x10^{-19}}$✓ OR $\frac{-1\ x\ 10^{-6}-(-6)\ x10^{-6}}{-1{,}6\ x10^{-19}}$
= 3,13 x 10^{13}✓ (3)

[15]

QUESTION 6

6.1.1 Single vector, ✓ which has the same effect as the original vectors acting together✓

(2)

6.1.2 Change ✓in position✓ (2)

6.1.3 R = $\sqrt{30^2 + 20^2}$✓ ✓sketch
= 36,06 m

Tan $\theta = \frac{30}{20}$ = 56,31^0✓. OR Tan $\alpha = \frac{20}{30}$ = 33,69^0. (3)

6.1.4 Speed = $\frac{20+30}{10+10}$ ✓= 2,5 m.s^{-1}✓ (2)

6.1.5 V = $\frac{36{,}1 \ coe \ Q \ 6.1.3 \ ✓}{20}$ = 1,805 m.s^{-1}✓ (2)

6.2 Vperson = $\frac{d}{t}$ ✓ Vlion = $\frac{d+26}{t}$

$t = \frac{d}{4}$ ✓ t = $\frac{d+26}{6}$✓

$\frac{d}{4} = \frac{d+26}{6}$✓

d = 52 m ✓ (5)

6.3.1 Rate of change✓ of displacement✓ (2)

6.3.2 AB – increasing velocity✓ right✓, constant acceleration✓

BC – constant velocity✓ (4)

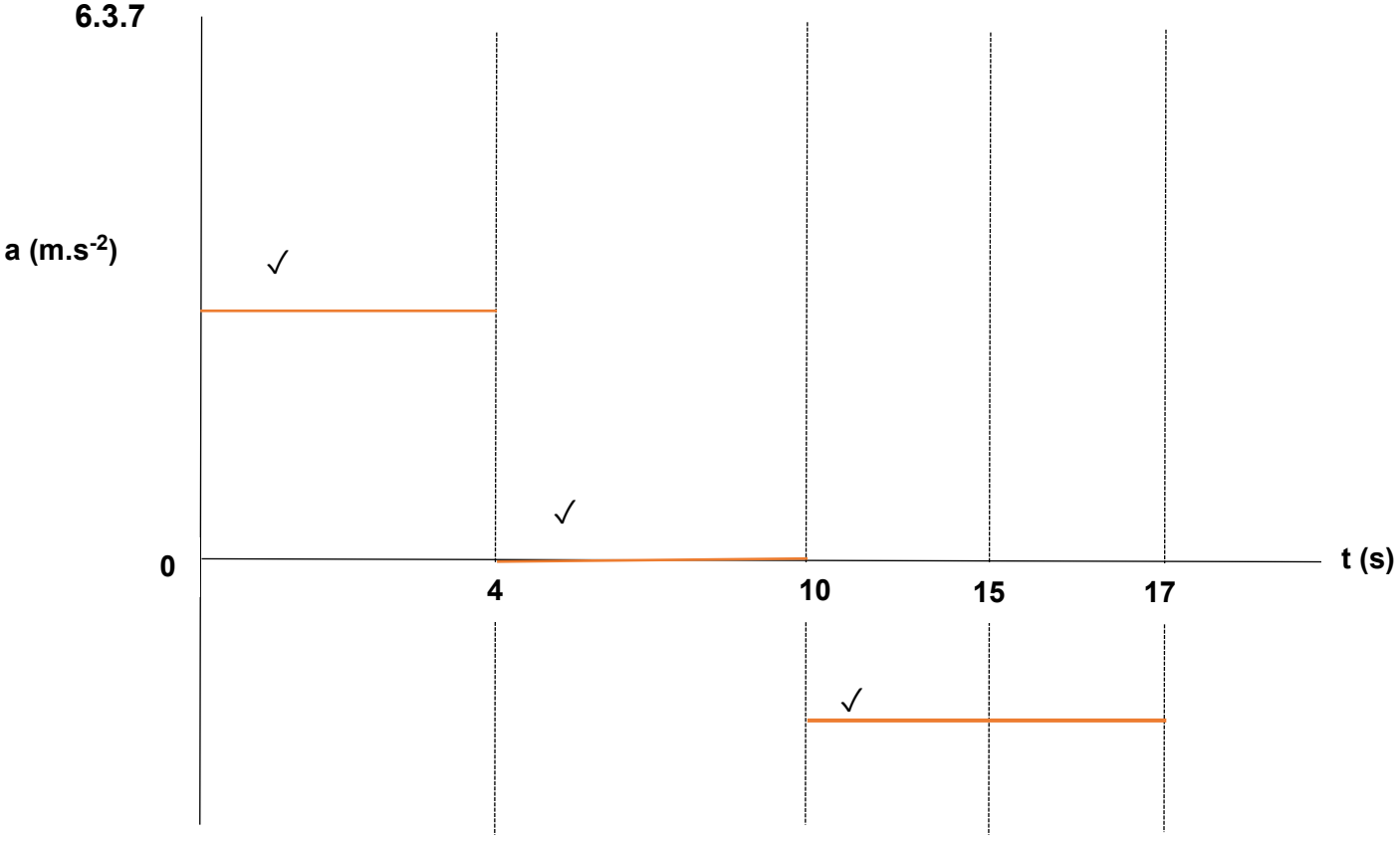
6.3.3 a = $\frac{12-0}{10-15}$✓
= -2,4
= 2,4 m.s^{-2} ✓left✓ (3)

6.3.4 Increasing velocity ✓left✓, constant acceleration✓ (3)

6.3.5 -2,4 (coe Q 6.3.3) = $\frac{x-0}{17-15}$✓
X = -4,8
= 4,8 m.s^{-1} ✓left✓ (3)

6.3.6 X = ½ (4)(12) + (6x12) + ½ (5)(12) ✓ – ½ (2(4,8 coe Q6.3.5) ✓
= 24 +72 +30 – 4,8
= 121,2 m ✓right✓ (4)

6.3.7

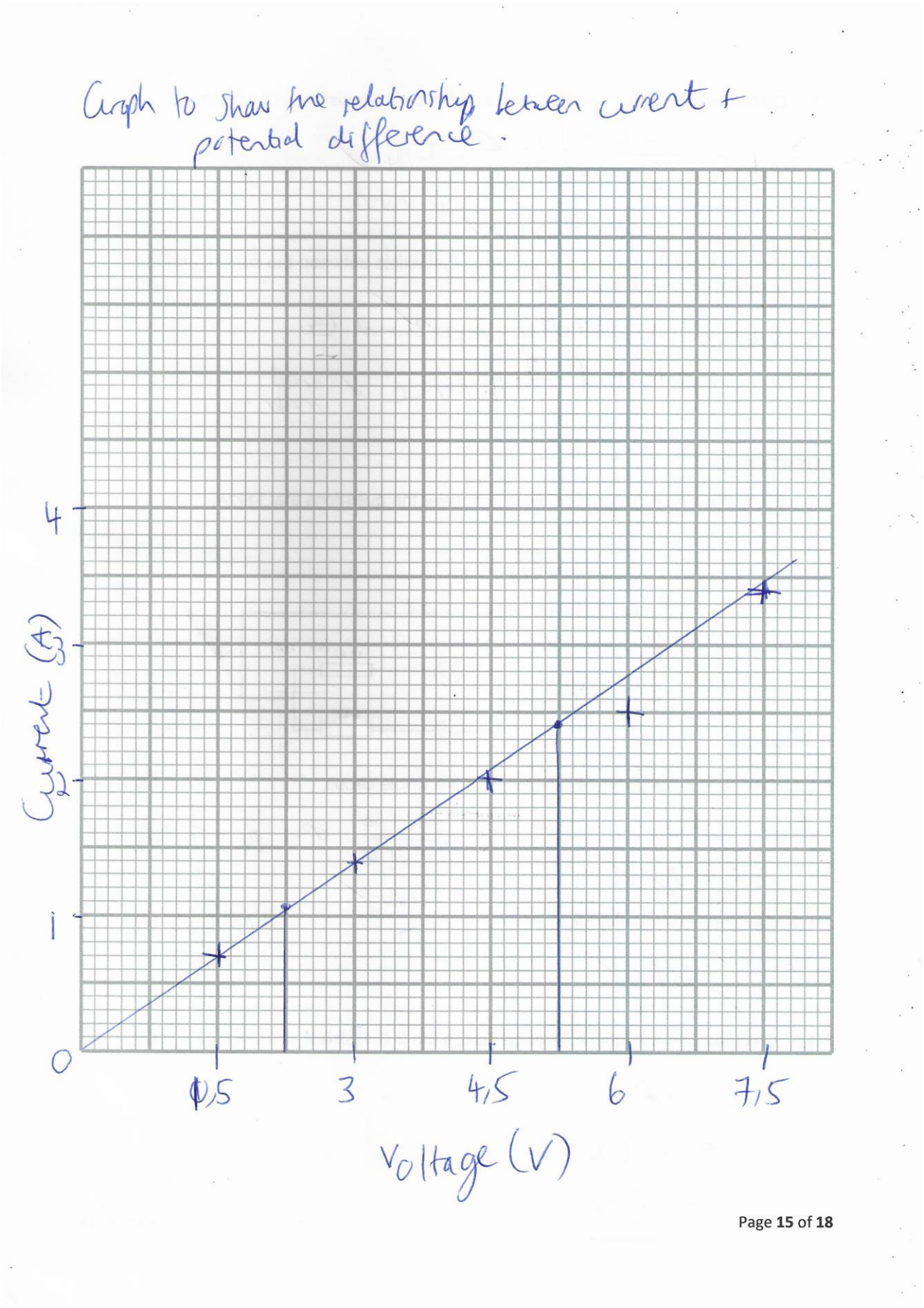


[38]

QUESTION 7

7.1 Current✓ (NOT ammeter) (1)

7.2 Temperature✓ OR same resistor or same battery (1)



scale 1, line of best fit 1, plotting 2
-1 if axes swopped

7.4 Gradient = $\frac{2,4-1,1}{5,25-1,75}$ ✓ ✓ (only one if use values from table)= 0,4✓

Invert R = 2,5 Ω✓ (4)

7.5 Current is directly proportional✓ to potential difference

Correct order✓ (2)

[15]

QUESTION 8

8.1.1 $\frac{1}{R} = \frac{1}{6} + \frac{1}{12}$ or prod/sum✓

$R = 4\Omega$

R = 4 + 2 ✓= 6 Ω✓ (3)

8.1.2 I = $\frac{V}{R}$✓

= $\frac{18}{6}$✓

= 3 A✓

8.1.3 V = IR

= (3)(2) ✓

= 6 V✓

8.1.4 V// = 18-6 =12V✓

$A_2 = \frac{V}{R}$✓

= $\frac{12}{12}$

= 1 A✓

$A_3 = \frac{V}{R}$

= $\frac{12}{6}$

= 2 A✓

OR Ratio method

8.2

8.2.1 Work done ✓or energy transferred per unit positive charge ✓ (2)

8.2.2 3 A✓ (1)

8.2.3 $I = \frac{Q}{t}$

$Q = It$✓

$= (3)(2x60)$ ✓

$= 360$ C✓ (3)

8.2.4 $V_2 = IR$

$= (3)(3)$ ✓

$=9$ V

$V_3 = 21-9$✓ OR $R = \frac{V}{I} = \frac{21}{3} = 7\Omega$ $Rp = 7-4 = 4\Omega$

$= 12$ V

$Rp = \frac{12}{3}$✓ Or 1,5 A in each branch $R = 12/1,5 = 8\ \Omega$

$= 4\ \Omega$

$\frac{1}{4} = \frac{1}{x} + \frac{1}{x}$✓

$X = 8\ \Omega$✓ (5)

[23]